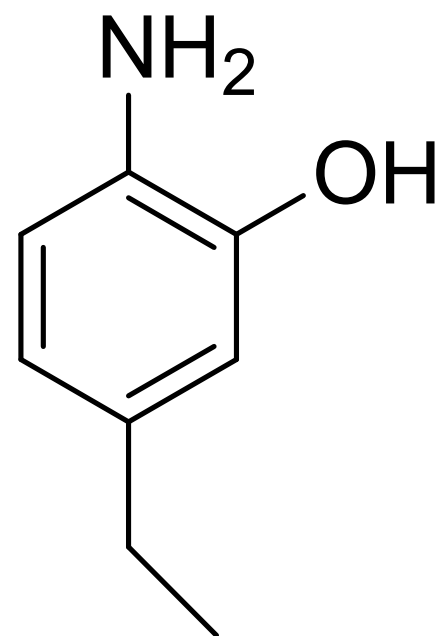
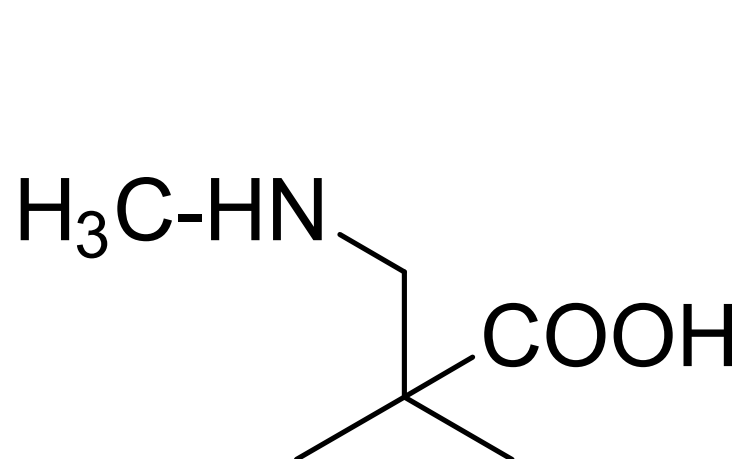
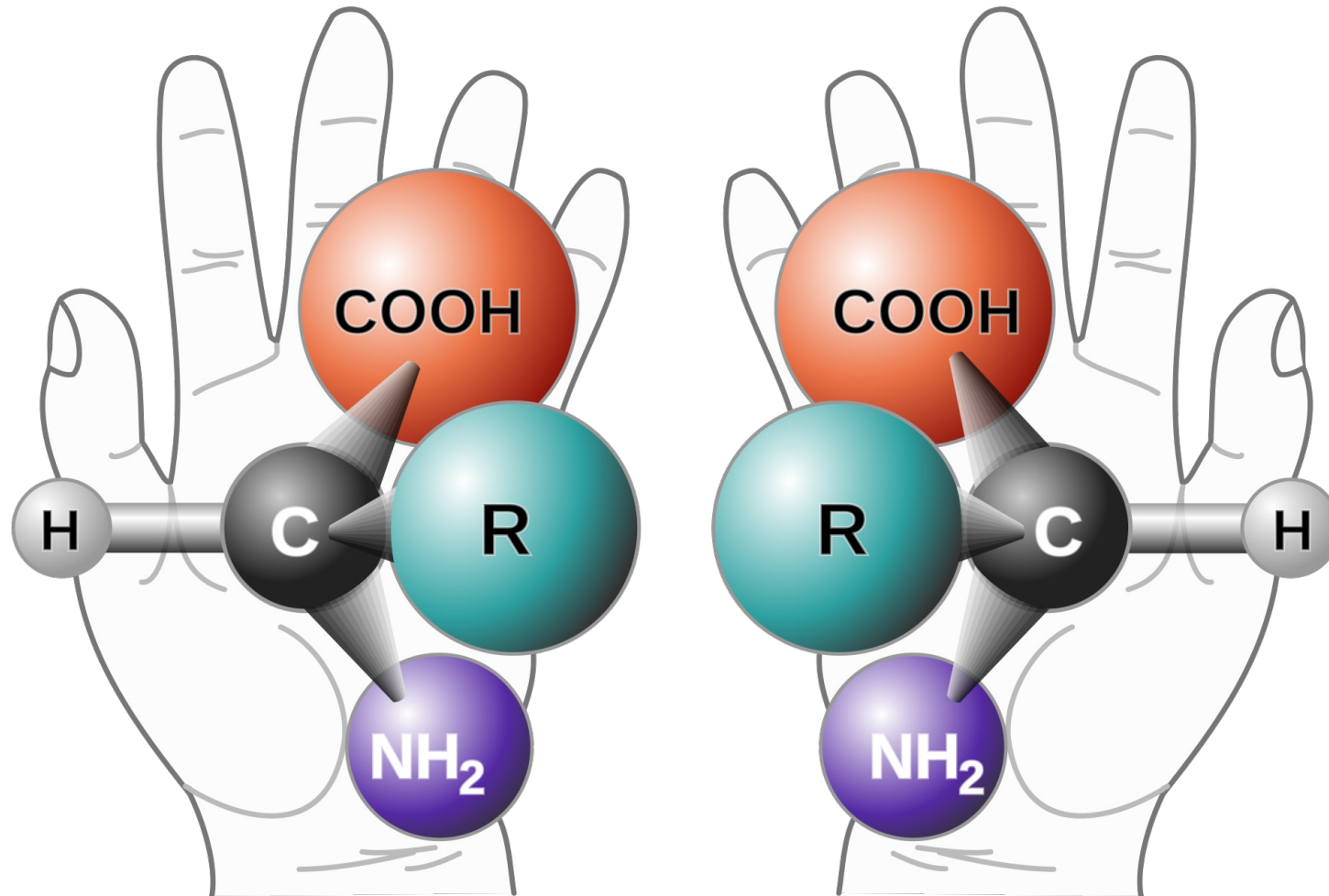


Task

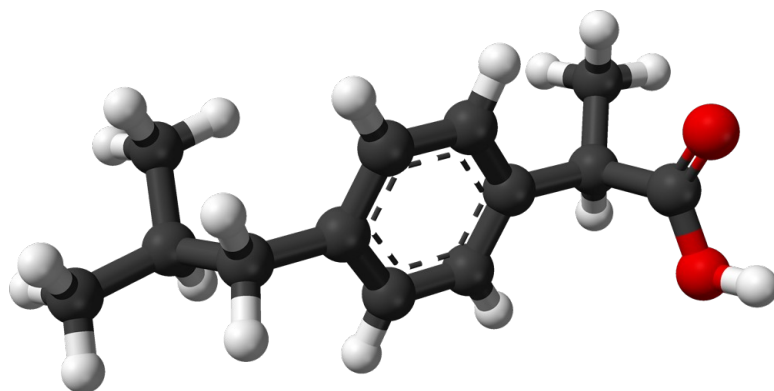
Identify functional groups in the structures below



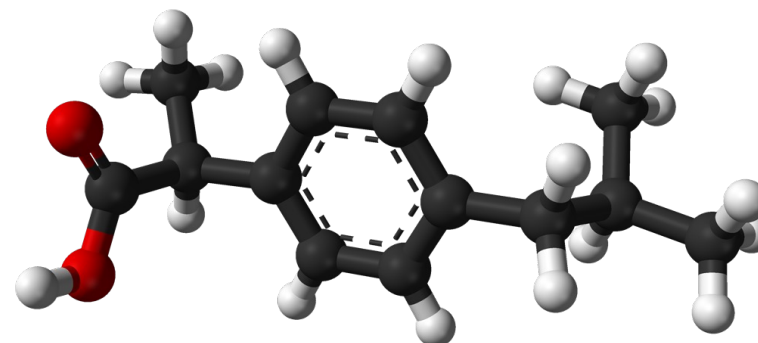
Chirality



Chiral molecules have C atoms with four different groups attached. Switching two of these will create an **enantiomeric pair**.
The asymmetric atom is labelled with *.



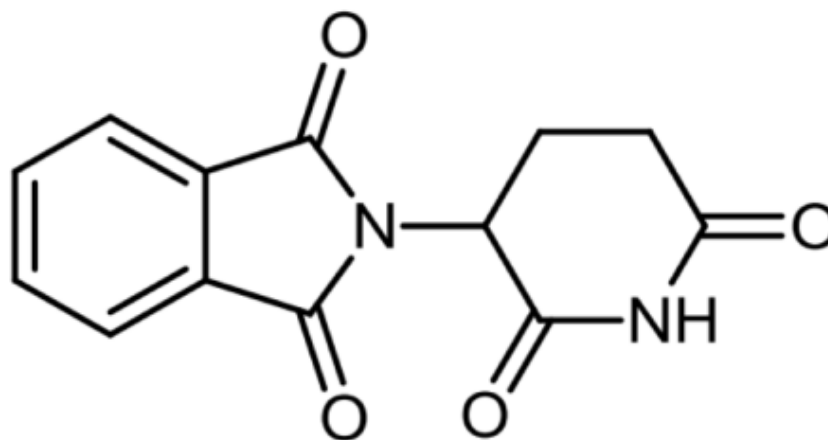
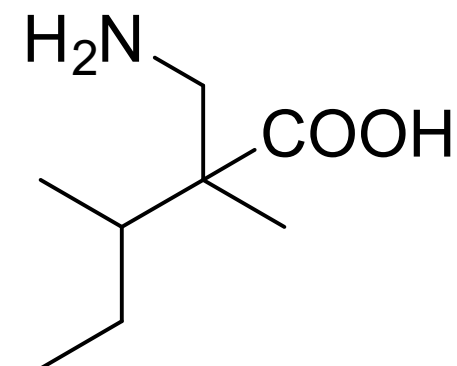
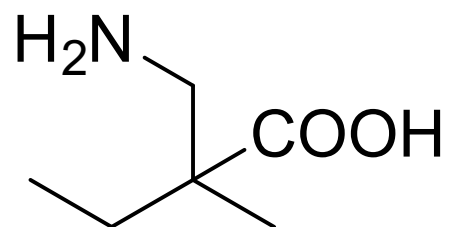
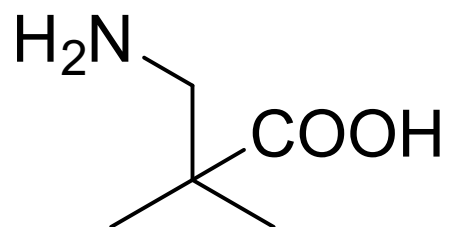
R-ibuprofen



S-ibuprofen (active form)

Exercise

Find chiral centers in the structures below:



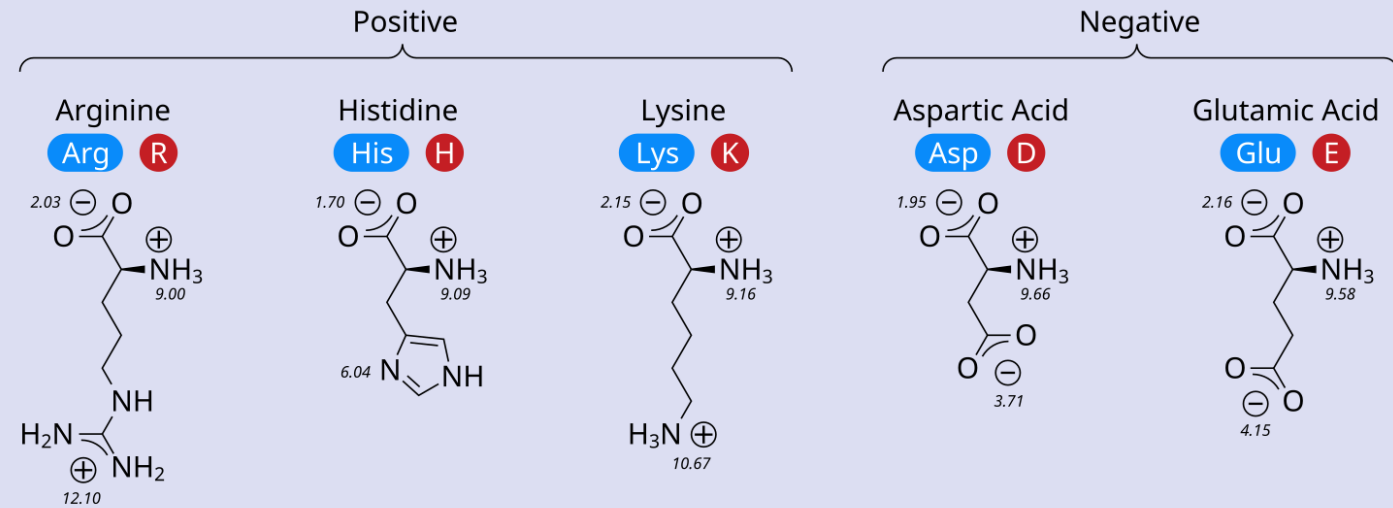
TWENTY-ONE PROTEINOGENIC α -AMINO ACIDS

Side chain charge
at physiological
pH 7.4

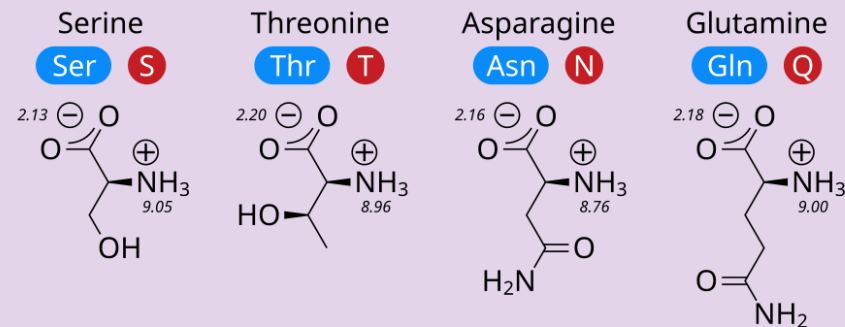
pK_a values shown
italicized

⊕ Positive
⊖ Negative

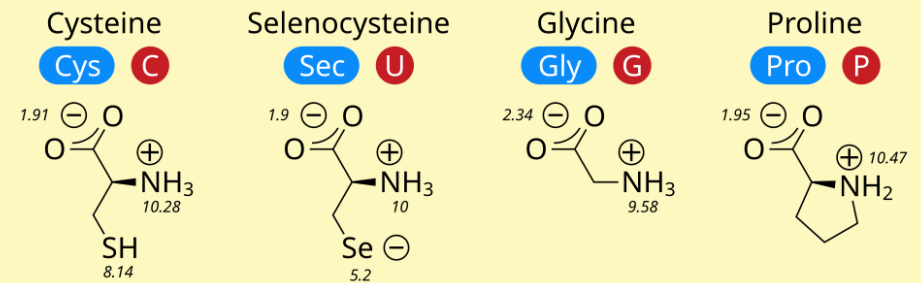
A. Amino Acids with Electrically Charged Side Chains



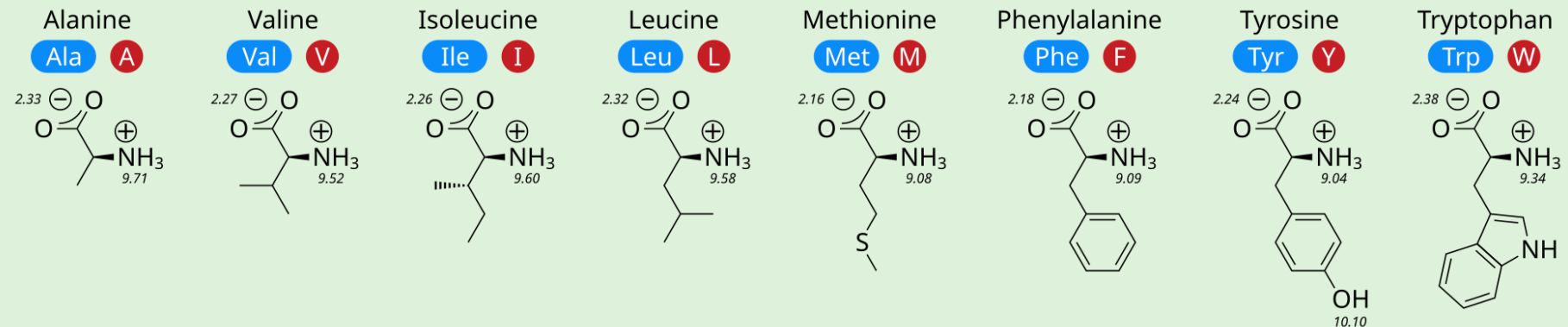
B. Amino Acids with Polar Uncharged Side Chains



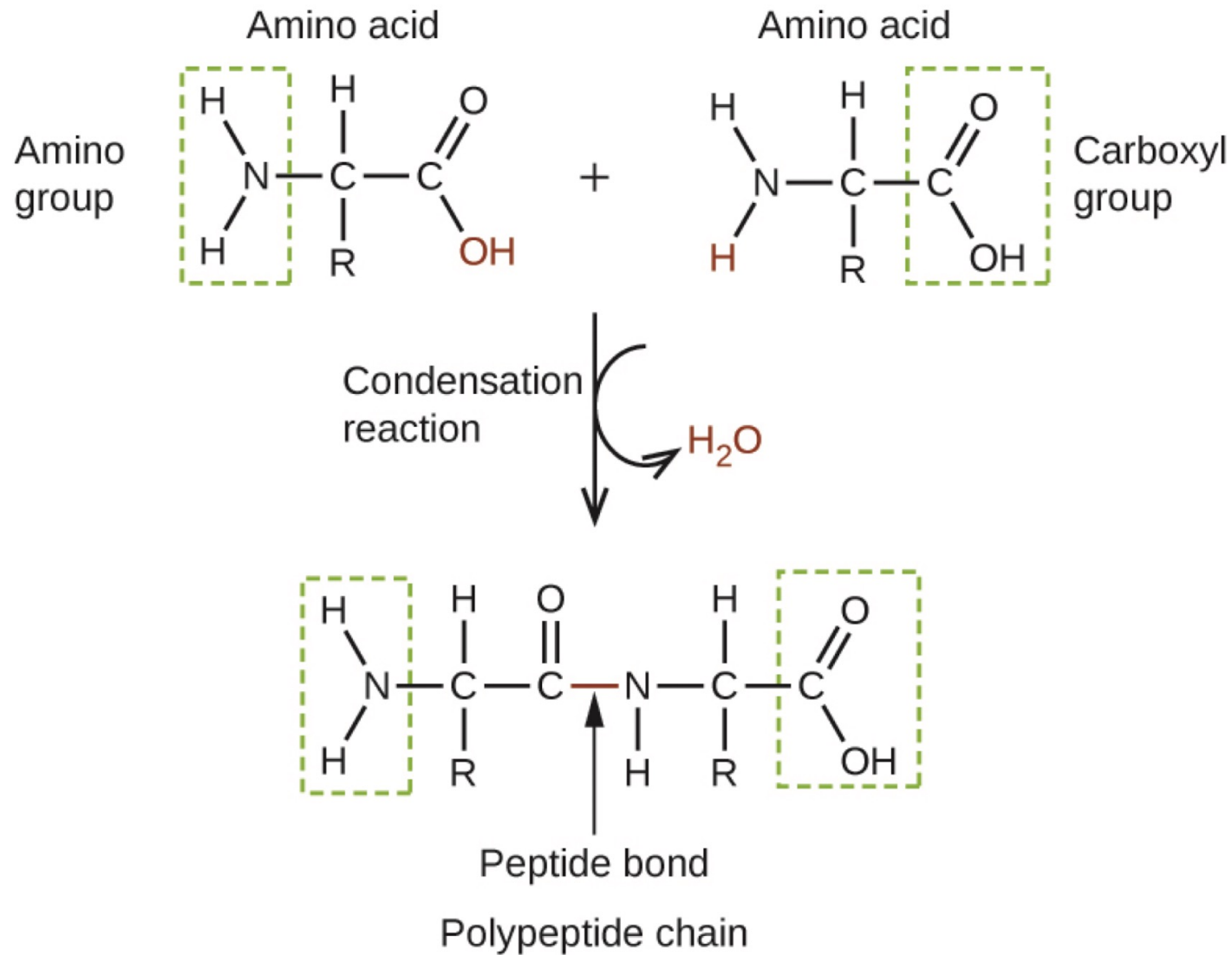
C. Special Cases



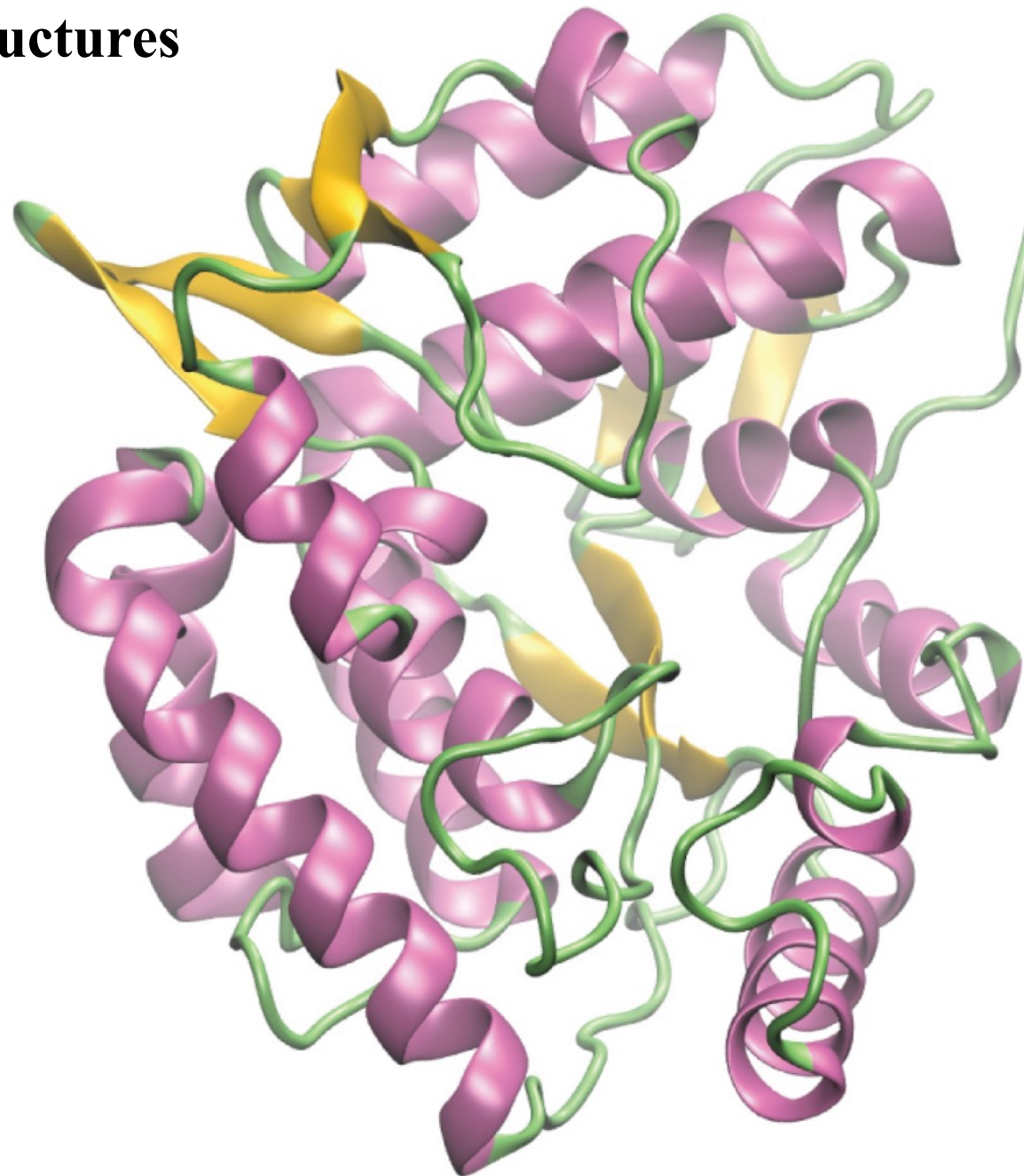
D. Amino Acids with Hydrophobic Side Chains



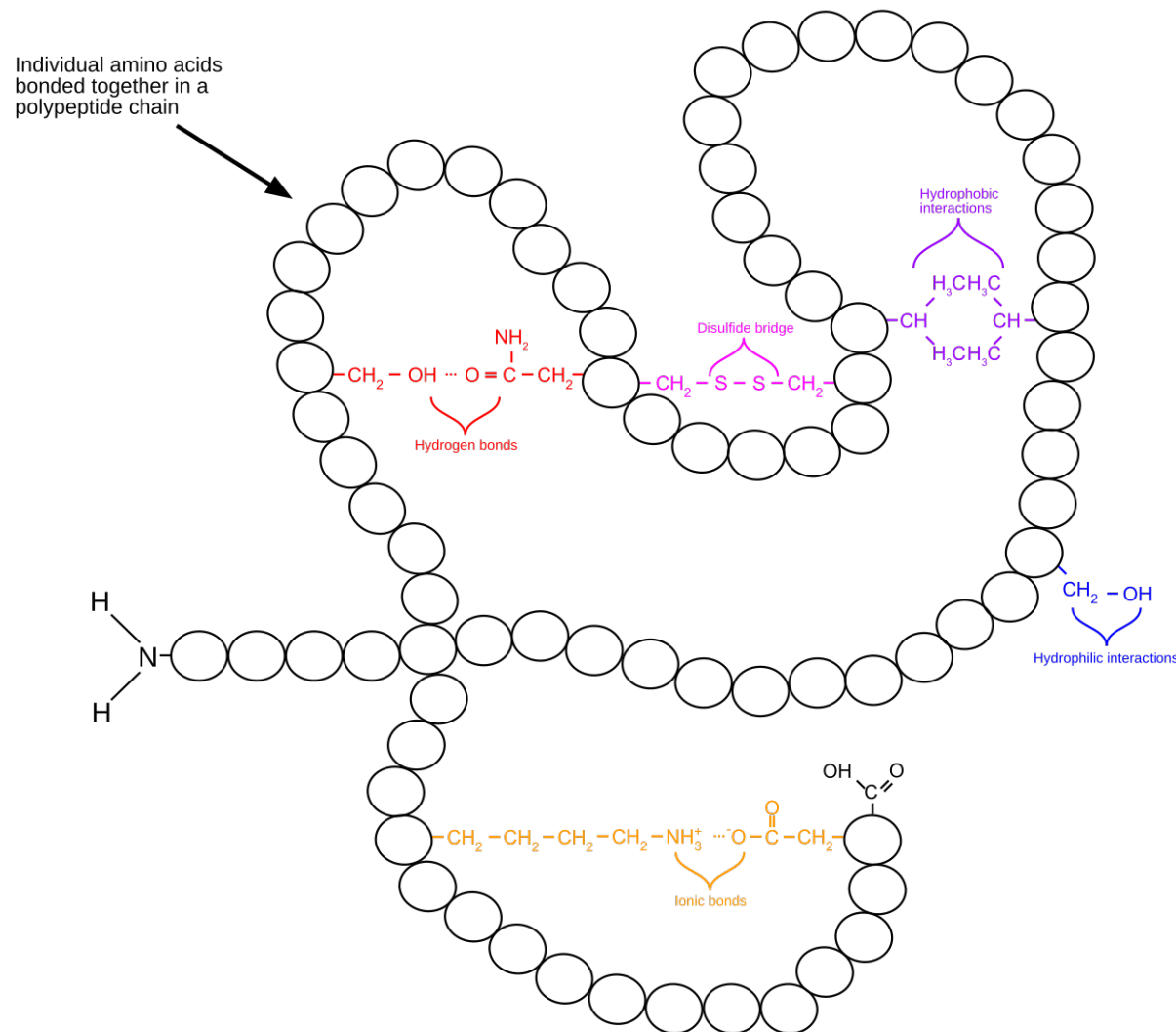
Amino acids form peptides and proteins



Protein structures



Protein structures



Bond types

Hydrophobic interactions:

These amino acids orient themselves towards the center of the polypeptide to avoid the water

Disulfide bridge: Cysteine forms a bond with another cysteine through its side group

Hydrogen bonds: Polar side groups on the amino acids form bonds with other polar side groups

Hydrophilic interactions: These amino acids orient themselves outward to be close to the water

Ionic bonds: Positively charged side groups bond to negatively charged ones

Acid and Bases & Solubility Products (Ch. 14+15)

26021 Chemistry (Polytechnical Foundation)

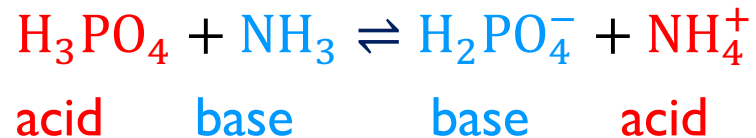
Professor Kasper S. Pedersen

2026

Brønsted's definition

Brønsted's definition of acids and bases:

An acid can dispose a proton
– a base can take up a proton



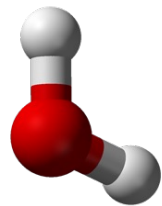
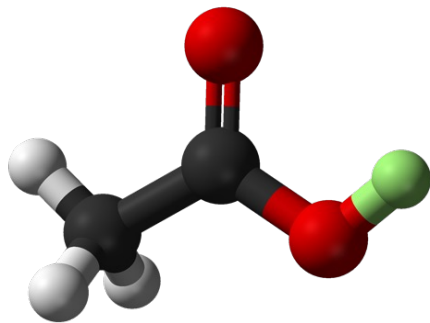
H_2PO_4^- is the corresponding (conjugate) base of H_3PO_4 .

NH_4^+ is the corresponding acid of NH_3

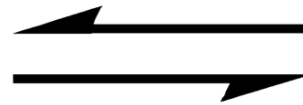


Brønsted's definition

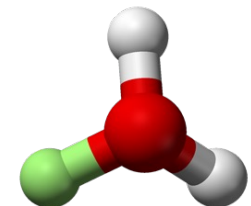
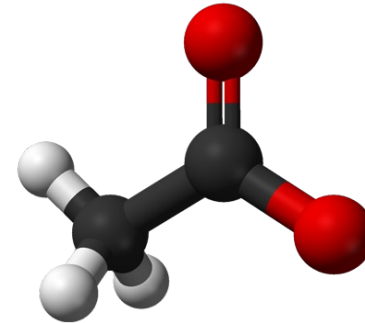
"Acetic acid"
 CH_3COOH



Water



"Acetate"
 CH_3COO^-



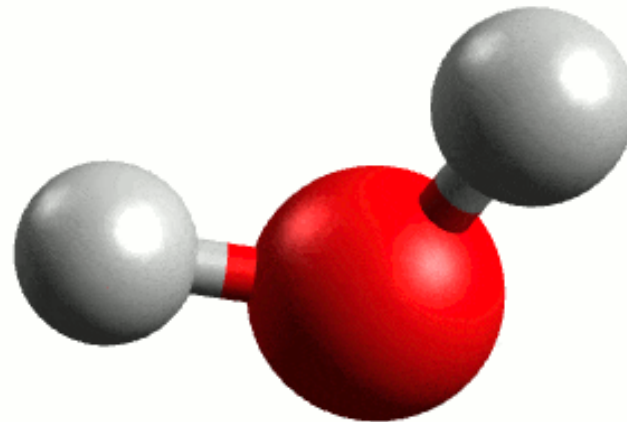
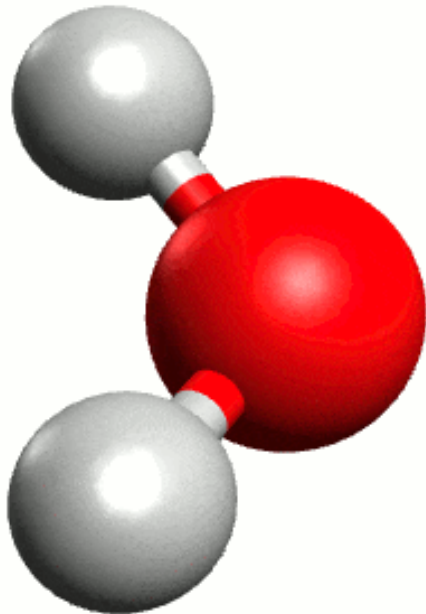
Oxonium
 H_3O^+

Acids and bases: water

- A compound - such as water can be both a base and an acid



base acid acid base



Acids and bases: water

A compound - such as water can be both a base and an acid



$$K_w = [\text{H}_3\text{O}^+] [\text{OH}^-] = 1.0 \times 10^{-14} \text{ M}^2 \text{ (at 298 K)}$$

In *neutral* solution: $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$

In *acidic* solution: $[\text{H}_3\text{O}^+] > [\text{OH}^-]$

In *basic* solution: $[\text{H}_3\text{O}^+] < [\text{OH}^-]$

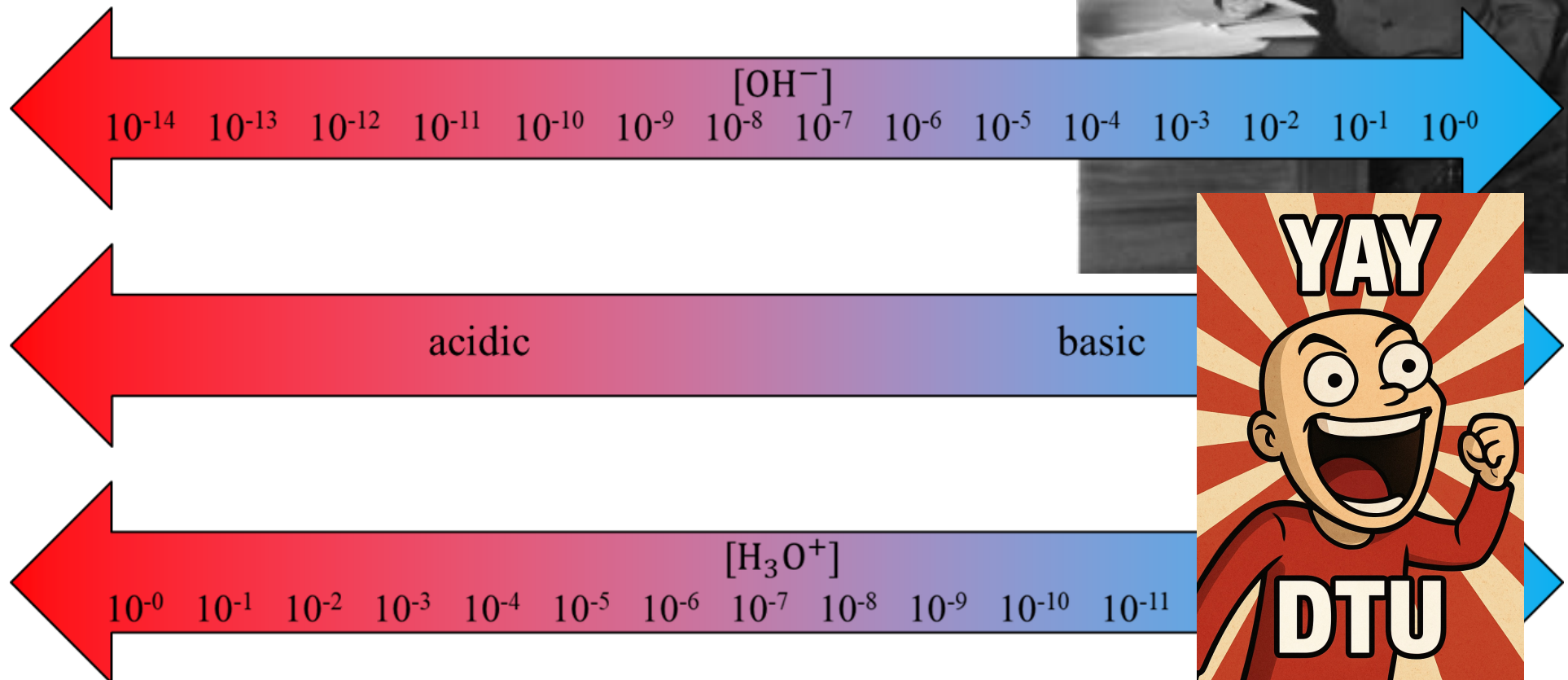
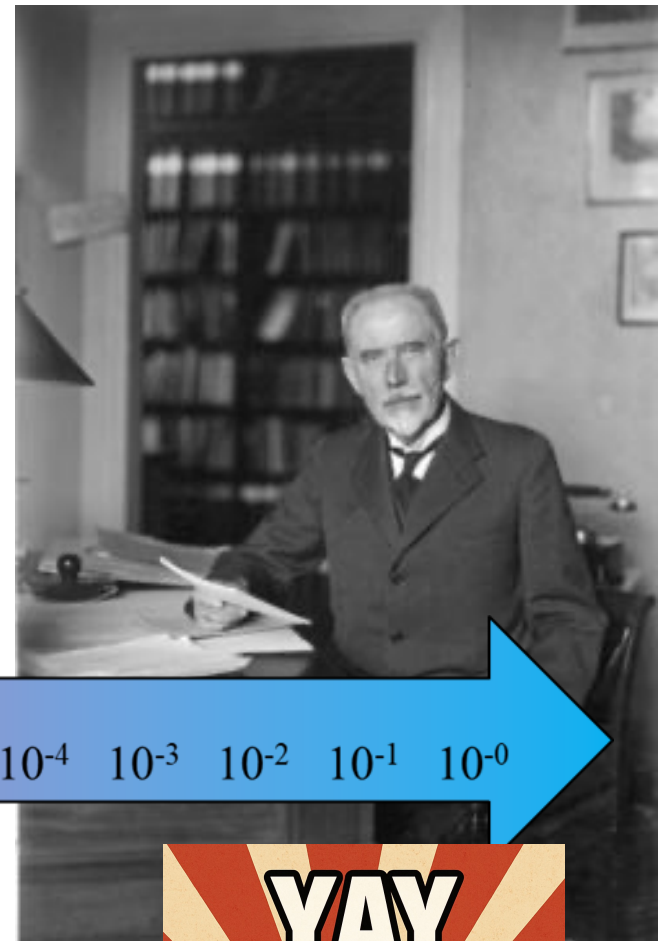
Example: Is a solution with $[\text{H}_3\text{O}^+] = 2.0 \times 10^{-4} \text{ M}$ basic or acidic?

$$[\text{OH}^-] = 1.0 \times 10^{-14} \text{ M}^2 / 2.0 \times 10^{-4} \text{ M} = 5.0 \times 10^{-11} \text{ M (acidic)}$$

The pH scale

Definition:

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

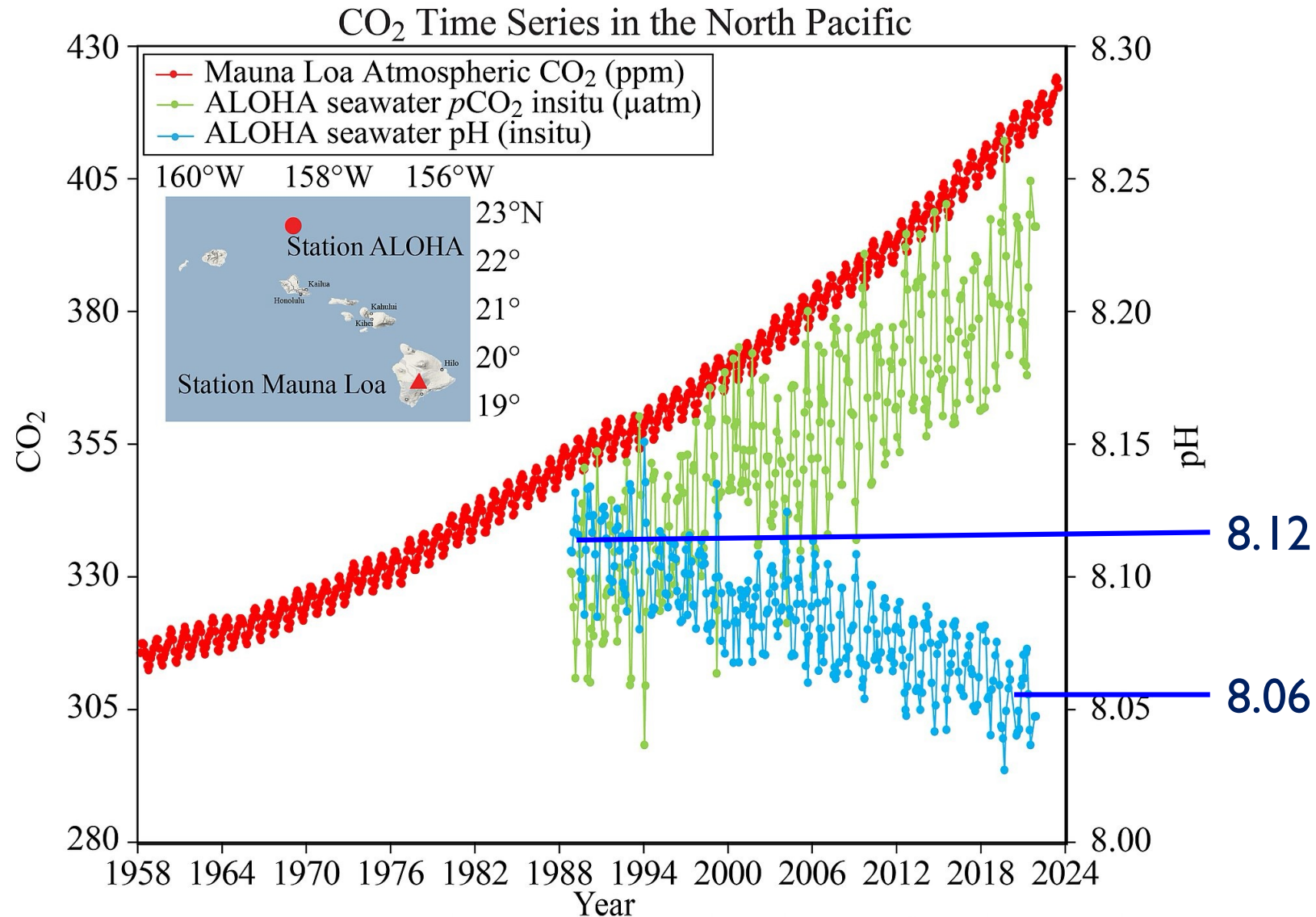


pHs of some common fluids

Sample	pH	Sample	pH
Gastric juice	1–2	Lemon juice	2.4
Vinegar	3.0	Grapefruit juice	3.2
Orange juice	3.5	Urine	4.8–7.5
Water exposed to air	5.5	Saliva	6.4–6.9
Milk	6.5	Pure water	7.0
Blood	7.35– 7.45	Tears	7.4
		Household ammonia	11.5

Exercise

Calculate the percentual increase in the oceans' $[\text{H}_3\text{O}^+]$ from 1988 to 2023.



Acid strength



HA: strong electrolyte (completely dissociated) $\Rightarrow$ *strong acid*

HA: weak electrolyte (not completely dissociated) $\Rightarrow$ *weak acid*

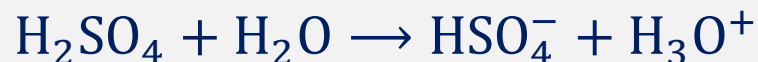
nitric acid:



hydrochloric acid:



sulfuric acid:

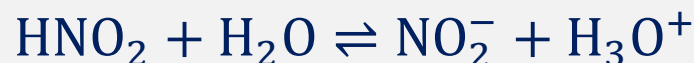


perchloric acid:

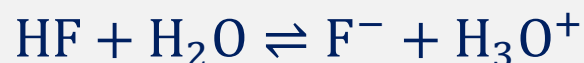


Examples of
strong acids

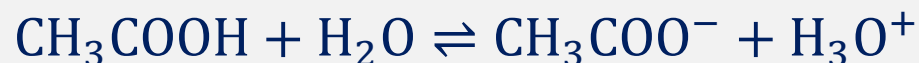
nitrous acid:



hydrofluoric acid



acetic acid



Examples of
weak acids

Acid constant

The general reaction:



has the equilibrium constant:

$$K_a = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]}$$

Defined as the *acid (ionization) constant* ("syrestyrkekonstant"), K_a .

As for 'pH', we can define

$$\text{p}K_a = -\log(K_a)$$

Base constant

Analogously, for a general reaction:



we have the equilibrium constant:

$$K_b = \frac{[\text{HB}][\text{OH}^-]}{[\text{B}^-]}$$

defined as the *base (ionization) constant* ("basestyrkekonstant"), K_b .

As for 'pH', we can define

$$\text{p}K_b = -\log(K_b)$$

For corresponding acid/base pairs:



Multiplication gives:

$$K_a(\text{HA})K_b(\text{A}^-) = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]} \frac{[\text{HA}][\text{OH}^-]}{[\text{A}^-]} = [\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 10^{-14} \text{ M}^2$$

Taking the logarithm on both sides yields:

$$pK_a(\text{HA}) + pK_b(\text{A}^-) = pK_w = 14$$

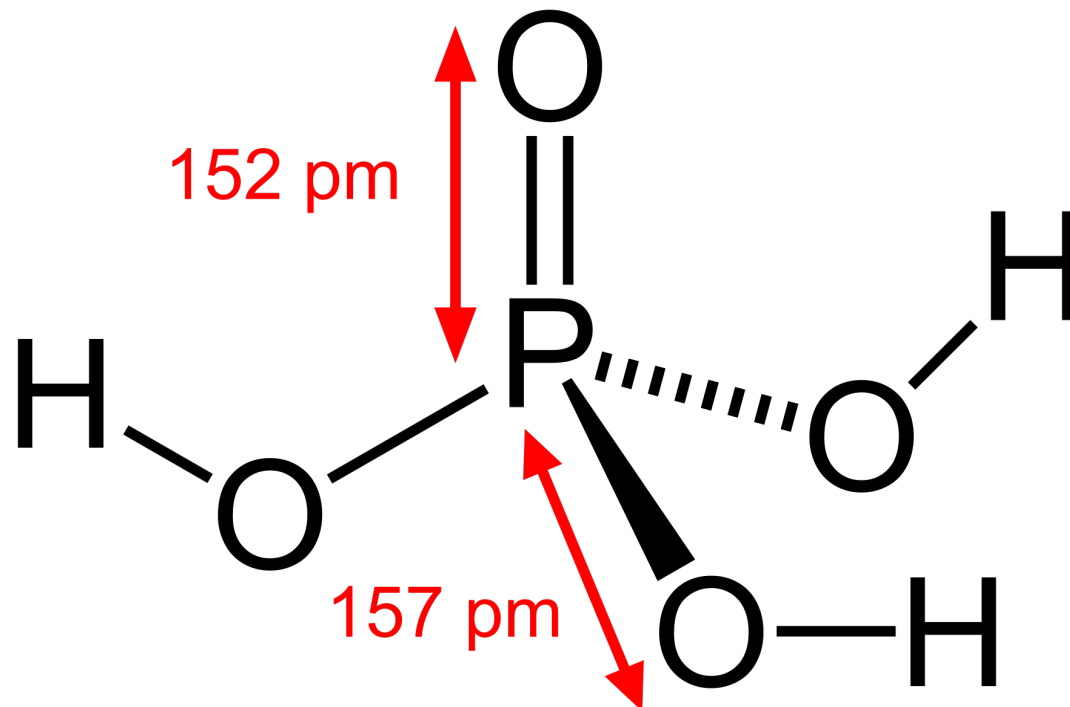
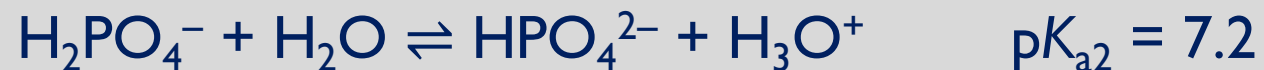
Strengths of Conjugate Acid-Base Pairs

Acid		Conjugate base	
HClO ₄	perchloric acid	ClO ₄ ⁻	perchlorate
HI	hydroiodic acid	I ⁻	iodide
HBr	hydrobromic acid	Br ⁻	bromide
HCl	hydrochloric acid	Cl ⁻	chloride
H ₂ SO ₄	sulfuric acid	HSO ₄ ⁻	hydrogen sulfate
HNO ₃	nitric acid	NO ₃ ⁻	nitrate ion
H ₃ O ⁺	oxonium	H ₂ O	water
HSO ₄ ⁻	hydrogen sulfate	SO ₄ ²⁻	sulfate
HF	hydrofluoric acid	F ⁻	fluoride
HNO ₂	nitrous acid	NO ₂ ⁻	nitrite
HCOOH	formic acid	HCOO ⁻	formate
CH ₃ COOH	acetic acid	CH ₃ COO ⁻	acetate
NH ₄ ⁺	ammonium	NH ₃	ammonia
HCN	hydrocyanic acid	CN ⁻	cyanide
H ₂ O	water	OH ⁻	hydroxide
NH ₃	ammonia	NH ₂ ⁻	amide

increasing acid strength ↑

↑ increasing base strength

Polyprotic acids: H_3PO_4



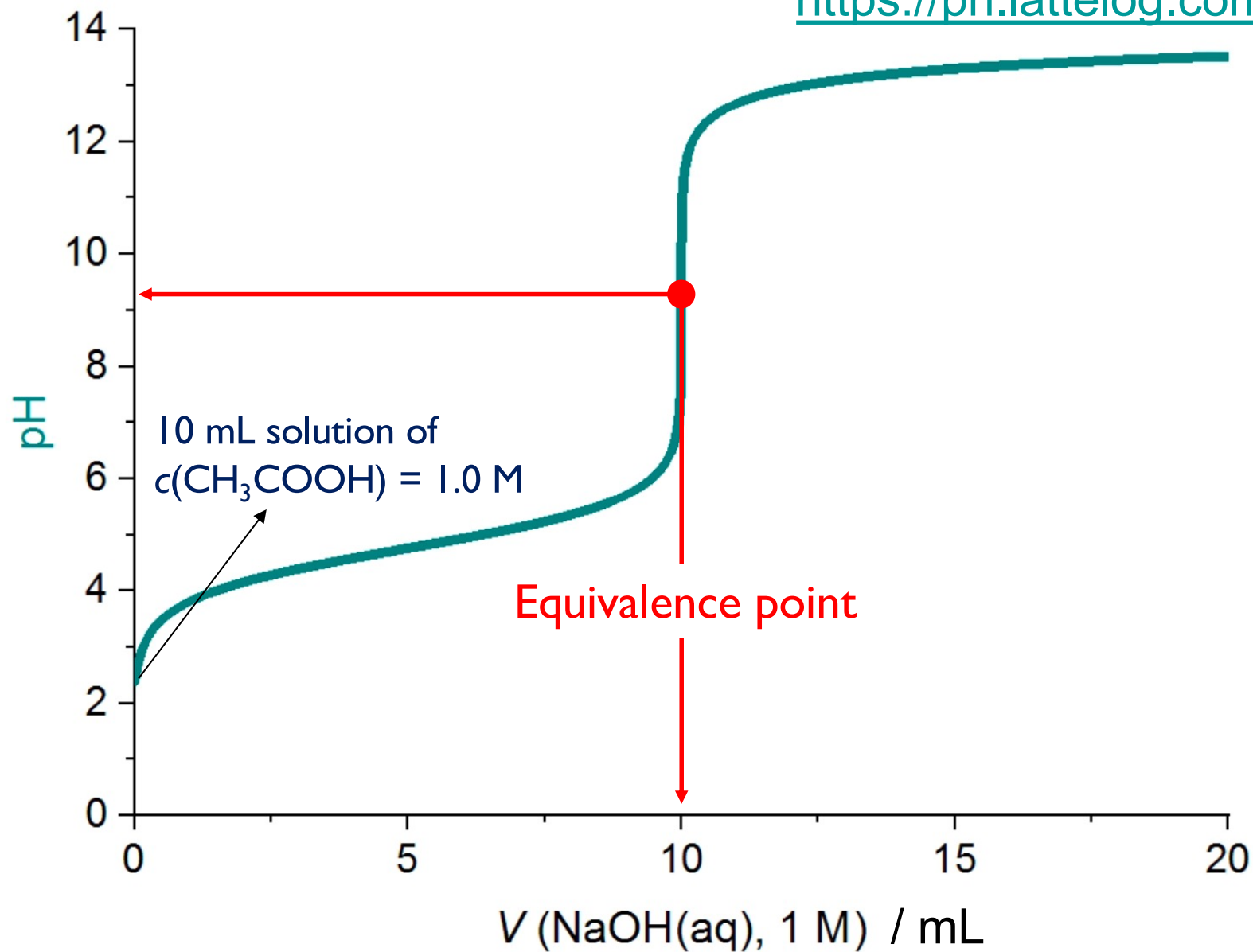
Exercise

Calculate the pH of:

- (a) an aqueous solution of HCl (strong acid, with $c(\text{HCl}) = 2.5 \times 10^{-3} \text{ M}$).
- (b) a solution with $c(\text{Ba}(\text{OH})_2) = 0.0450 \text{ M}$.
- (c) a 0.036 M solution of nitrous acid, $\text{HNO}_2(\text{aq})$ ($\text{p}K_a = 3.35$).

Titration of CH_3COOH

<https://ph.lattelog.com/titrage>



Buffers

- (a) Calculate the pH of a 1.0 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$ M)
- (b) Calculate the pH of a 1.0 M CH_3COONa
- (c) Calculate the pH of a buffer system containing $[\text{CH}_3\text{COOH}] = 1.0$ M and $[\text{CH}_3\text{COO}^-] = 1.0$ M.
- (d) What is the pH of the solution after the addition of 0.10 mol gaseous HCl to 1.0 L of the solution (assuming no change in volume). Compare the result to the expected pH if 0.1 mol gaseous HCl is dissolved in 1.0 L pure water.

Preparing a buffer with a specific pH

The expression for the acid ionization constant reads:

$$K_a(\text{HA}) = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]} \quad \text{and therefore} \quad [\text{H}_3\text{O}^+] = \frac{K_a[\text{HA}]}{[\text{A}^-]}$$

Taking the logarithm yields

$$-\log[\text{H}_3\text{O}^+] = -\log \frac{K_a[\text{HA}]}{[\text{A}^-]} = -\log K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

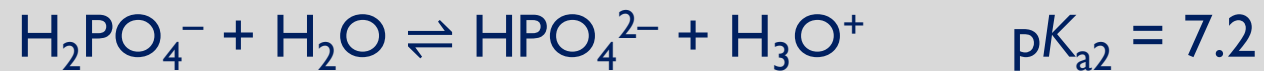
Which is equivalent to

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]} \quad (\text{note the fraction and sign inversion})$$

Henderson-Hasselbalch equation ("bufferligningen")

Exercise

How would you prepare a buffer with a pH of about 7.45 using "phosphates":

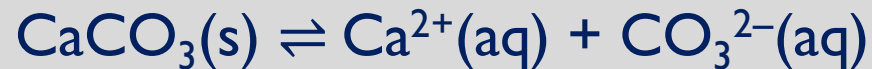


Solubility and solubility products



Solubility products

Example: Calcium carbonate is very slightly soluble in water and dissolves according to:



with the equilibrium constant (*the solubility product*)

$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{CO}_3^{2-}] = 3.3 \times 10^{-9} \text{ M}^2$$

Exercise:

- Calculate the solubility (g/L) of CaCO_3 in pure water.
- In healthy coral reef areas, $[\text{CO}_3^{2-}] \approx 10^{-4} \text{ M}$ and $[\text{Ca}^{2+}] \approx 10^{-2} \text{ M}$.
Comment.
- In which direction would the above equilibrium shift during ocean acidification?